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In Vitro Flooding of Childhood Posttraumatic Stress Disorders: A Systematic Replication

Philip A. Saigh
Graduate School and University Center
City University of New York

An *in vitro* flooding package was used to treat the posttraumatic stress disorders of three children. Traumatic scenes were identified and stimulus-response imagery cues were provided according to a multiple baseline design. The outcome data revealed positive changes on the children's affective, behavioral, and cognitive parameters. The relevance of the flooding modality is considered from within the scope of the practical needs of school psychology.

According to the American Psychiatric Association's (APA) *Diagnostic and Statistical Manual of Mental Disorders* (*DSM-III*; APA, 1980) posttraumatic stress disorders (PTSD) are precipitated by a distinct stressor that is "generally outside the range of usual human experience" and of sufficient intensity as to "invoke significant symptoms of distress from almost everyone" (p. 238). These symptoms are characterized by high levels of anxiety, unwanted recollections of the trauma, blunted affect, avoidance behaviors, and a number of disparate symptoms (e.g., concentration and memory impairment, survival guilt, and startle response). The *DSM-III* also stipulates that symptoms of depression as well as "sporadic and unpredictable explosions of aggressive behavior" may be evident and that "the disorder can occur at any age including childhood" (APA, 1980, p. 237). Within this context, children have developed PTSD after being abducted (Terr, 1983), physically abused (Blank & Dickerson, 1982), attacked by animals (Gislason & Call, 1982), raped (Kilpatrick, Veronen, & Best, 1985), or exposed to nuclear accidents (Collins, Baum, & Singer, 1983).

Examined from a behavioral paradigm, five single-case investigations used *in vitro* flooding (i.e., a behavioral treatment based on the principal of extinction and involving prolonged imaginal exposures to highly aversive stimuli) (O'Leary & Wilson, 1987) to treat circumscribed PTSD cases. Three of these reports (Fairbank, Gross, & Keane, 1983; Fairbank & Keane, 1982; Keane & Kaloupek, 1982) involved the treatment of American combat veterans. The fourth and fifth reports (Saigh, 1985a, in press) are more germane to the practice of school psychology inasmuch as they involved the treatment of traumatized students. Within this context, Saigh's reports used multiple-component behavioral assessments (i.e., behavioral avoidance tests, self-reported measures of anxiety and depression, and self-monitored frequencies of traumatic ideation) as well as measures of concentration and memory impairment to assess the varied symptomotology of PTSD before, during, and after *in vitro* flooding. The results clearly

Requests for reprints should be sent to Philip A. Saigh, Department of Educational Psychology, Graduate School and University Center, City University of New York, 33 West 42nd Street, New York, NY 10036.

revealed that the flooding process had a positive influence on the affective, behavioral, and cognitive outcome measures of a 14-year-old Lebanese boy who had been abducted and tortured (Saigh, in press) and a 9-year-old Lebanese girl who had been traumatized during a shelling incident (Saigh, 1985a).

Although the extant *in vitro* flooding results are promising, information pertaining to the efficacy of the flooding modality with school-age children is limited to Saigh's (1985a, in press) single-case reports. With this in mind and in view of the debilitating effects of the disorder, this study reflects a systematic replication (Barlow & Hersen, 1984) of Saigh's earlier studies. More specifically, it examines the affective, behavioral, and cognitive parameters of the disorder throughout the *in vitro* flooding of three traumatized children.

METHOD

Cases

Each case was assessed and treated by the author at a clinical setting in Lebanon. The first case (Layla) involved an 11-year-old Lebanese girl who was referred for evaluation by her school principal because of poor grades and a chronic level of disruptive behavior (e.g., talking out, consistent refusals to comply with her teachers' directions, and temper outbursts). Nonstructured interviews with Layla, her parents, principal, and teacher revealed that these problems developed shortly after she witnessed the death of a man by a sniper. Layla presented 32 months after the incident with trauma-related complaints involving anxiety-evoking recollections of the trauma. To a lesser extent she complained of survival guilt, blunted affect, avoidance behaviors, and concentration impairment. Her parents corroborated this account and added that her symptoms had not remitted.

The second case (Sawsan) concerned an 11-year-old Lebanese girl who was referred for assessment by her teacher. Nonstructured interviews with Sawsan, her parents, and teacher revealed that Sawsan saw two pedestrians die during a shelling incident. Sawsan presented 36 months after the incident with intrusive anxiety-evoking recollections of the trauma as well as trauma-related nightmares. She also said that she was less interested in visiting friends and that she felt depressed. To a lesser extent, Sawsan complained of survival guilt, memory and concentration impairment, and trauma-related avoidance behaviors. Her teacher indicated that Sawsan's limited attention span had had a negative influence on her academic achievement and that she frequently quarrelled with her classmates. Her teacher also indicated that Sawsan appeared to be overly anxious and sensitive to criticism. Sawsan's parents agreed with these comments and reported that their daughter's symptoms had not remitted since the time of the trauma.

The third case (Emile) involved a 12-year-old Lebanese boy who was referred for evaluation by his parents who revealed during nonstructured interview that the family home had been partially destroyed during a shelling incident. Emile presented 28 months after the incident with primary complaints involving anxiety-evoking recollections of the trauma. Secondary complaints involving trauma-related avoidance behaviors, depression, startle response, irritability, concentration impairment, and academic difficulties were also expressed. His principal was subsequently interviewed and it was indicated that Emile had become "quarrelsome" since the shelling incident and that his grades were well below average. Emile's parents concurred with these comments and voiced concern about the quarrels that Emile had been having at home and at school.

Each family was informed independently about the symptomatology and course of PTSD. Information was also provided about the process, duration, and efficacy of *in* *vitro* flooding and systematic desensitization (cf. Rachman & Wilson, 1980). The families went on to confer privately, and it was unanimously decided to pursue a course of *in vitro* flooding. The decisions to pursue the flooding regimen were based on three considerations. The first involved the positive results that were observed when *in vitro* flooding was used to treat the aforementioned adult, adolescent, and childhood PTSD cases. The second consideration was based on the suggestion that *in vitro* flooding may entail fewer treatment sessions than systematic desensitization (Rychtarik, Silverman, Van Landingham, & Prue, 1984). The final consideration was based on the author's clinical records that reflect three unsuccessful attempts to treat PTSD through systematic desensitization.

Traumatic Scenes

Each student was interviewed and chronological sequences of trauma-related anxiety-evoking scenes were established. Table 1 reflects the basic components of these scenes.

Measures

In an effort to gauge the affective, behavioral, and cognitive parameters of PTSD, a multiple-component assessment package was presented.

Affective measures. Initially, the Revised Children's Manifest Anxiety Scale (RCMAS; Reynolds & Richmond, 1978) and the Children's Depression Inventory (CDI; Kovacs, 1978) were administered. The RCMAS presents 39 anxiety-related items (e.g., "I am afraid of a lot of things") that are marked on a true or false basis. Each true response is scored as a 1 and each false response is scored as a 0. As such, RCMAS scores may range from 0 to 39. The CDI consists of 27 depression-related items (e.g., "I am sad once in awhile. I am sad many times. I am sad all the time"). A three-alternative, forced-choice format is used and each item is scored on a 0–2 basis resulting in a possible range of scores from 0 to 54. The RCMAS and CDI have enjoyed considerable utility in clinical, educational, and research settings since their development (Kazdin, 1983; Reynolds & Paget, 1981).

The incidence of unwanted and intrusive trauma-related thoughts were self-monitored on pocket frequency counters (i.e., the Knit Taly, Boyle Needle Company).

TABLE 1
Traumatic Scenes

Case	*Scene Number*	*Content*
Layla	1	Walking to neighborhood market
	2	Hearing shouts and seeing people flee
	3	Seeing a man's prostrate body in the street
	4	Running away
Sawsan	1	Standing by a window and hearing distant explosions
	2	Hearing the horns of cars in the street below and looking down
	3	Hearing a loud explosion
	4	Seeing the bodies of two pedestrians in the street
Emile	1	Hearing progressively closer explosions
	2	Seeking shelter in the building's staircase
	3	Hearing a very loud explosion
	4	Seeing the wreckage of his room

Finally, each student's level of anxiety vis-à-vis the respective traumatic scene sequences was measured through subjective units of disturbance (SUDS) ratings ranging from *no discomfort* (0) to *maximum discomfort* (10).

Behavioral measures. Individualized 10-item behavioral avoidance tests (BATs) were developed in order to evaluate the extent of each student's avoidance behaviors. Layla's BATs called for walking to the market where the man had been shot, standing by the place where the body had been, and walking home along the route that had been followed at the time of the trauma. Sawsan's BATs involved walking to the intersection where the shell had impacted and standing in the area where the bodies had fallen. Emile's BATs involved taking a seat in the room that had been shelled and listening to a recording of progressively closer explosions. It should be noted in this regard that Layla, Sawsan, and Emile's parents reported that their children had avoided the aforementioned places or activities since they had been traumatized. It should also be noted that each BAT had a time component (i.e., the behaviors had to be maintained for 30 min).

The student's classroom conduct was evaluated by their teacher against the Conners Teacher Rating Scale (CTRS; Conners, 1969) criteria. Briefly, the CTRS consists of 39 items (e.g., "restless and overactive") that are rated on a 4-point scale. As such, CTRS scores may range from 39 to 156. The CTRS has enjoyed considerable utility as a measure of behavioral disturbance since it was developed by Conners in 1969 (O'Leary, Vivian, & Nisi, 1985).

Cognitive measures. Initially, the WISC-R Digit Span and Coding subtests were administered. The Digit Span was administered because investigators have reported that it reflects short-term memory and because of its sensitivity of anxiety (Glasser & Zimmerman, 1967; Witmer, Bernstein, & Dunham, 1971). The Coding subtest was selected because of the suggestion in the literature that it measures concentration and freedom from distractibility (Kaufman, 1979). Finally, the student's trimester grade point averages (GPAs) were obtained from their school records. Information pertaining to the reliability and validity of the RCMAS, CDI, CTRS, and WISC-R as based on the responses of Lebanese students has been reported elsewhere (Saigh, 1985b, 1985d).

Research Design

Given that sequential withdrawal designs (e.g., A-B-A-B) are inappropriate when treatment variables cannot be withdrawn or reversed because of ethical or functional constraints (Barlow & Hersen, 1984), a multiple baseline across traumatic scenes design (Fairbank & Keane, 1982) was used. Within this framework SUDS levels served as the dependent variable. Viewed operationally, baseline records of each student's SUDS level were independently obtained for every traumatic scene before the *in vitro* flooding treatment was applied. Scene 1 was subsequently flooded (i.e., treated) and ongoing SUDS ratings for Scene 1 were taken throughout the flooding process. Immediately after the flooding session the remaining (i.e., untreated) scenes were briefly presented and SUDS ratings were obtained. During subsequent treatment sessions the remaining scenes were sequentially treated and SUDS levels were monitored. Following each of these treatment sessions SUDS ratings for the previously treated as well as the remaining or untreated scenes were obtained through probe assessments. This basic regimen was continued until all of the scenes had been treated. SUDS assessments for all of the traumatic scenes were subsequently taken during a posttreatment session 3 days after the last flooding session and during a follow-up session 6 months later.

Viewed conceptually, the baseline, treatment and follow-up assessments for each scene may be viewed as separate A-B-A-A designs with the A phase further extended for each of the succeeding scenes until the treatment was applied. Barlow and Hersen (1984) observed in this context that "the experimenter is assured that the treatment variable is effective when a change in rate of the concurrent (untreated) behaviors remains relatively constant" (p. 211).

The students' self-monitored index of traumatic thoughts was initiated 1 week before the multiple baseline SUDS probes were presented and continued on a daily basis throughout the baseline SUDS assessments and subsequent *in vitro* flooding. Self-monitoring was maintained for 7 consecutive days after the last treatment session and for 7 consecutive days 6 months later. The RCMAS, CDI, BAT, CTRS, and WISC-R subtests were administered according to a pretreatment, posttreatment, and follow-up regimen. More specifically, these measures were administered 1 week before the initial baseline session, immediately after the last treatment session, and 6 months later. The GPAs were indicative of the trimester grades that were recorded 2 weeks before the baseline probes, 10 weeks after the last treatment session, and 6 months after the last treatment session.

Multiple Baseline Procedure

Baseline probes. Baseline SUDS assessments were initiated by 10 min of therapist-controlled deep muscle relaxation exercises. Each of the traumatic scenes was subsequently presented for approximately 6 min. Stimulus response imagery cues (Fairbank & Keane, 1982; Levis, 1980) were used during each traumatic scene presentation. Stimulus cues involved the visual, auditory, olfactory, and tactile components of each scene. For example, Emile's stimulus cues involved the sounds of explosions, the image of a burning room, the smell of smoke, and the sensation of the heat that was emitted by the flames. Response cues involved the auditory, behavioral, and cognitive aspects of each scene (e.g., hearing expressions of panic, running out of the apartment, and catastrophic ideas involving the possibility of additional explosions). In this context, Layla and Sawsan were able to imagine and maintain the components of their scenes. Although Emile had no difficulty with Scene 4, he could not imagine the components of the former scene for more than a minute. In order to overcome this problem, the sounds of progressively closer shell and rocket explosions were recorded on a Realistic dual pattern 600 ohms microphone (Model 33–1062) and on a Realistic cassette recorder (Model 14–1024). The sound track was amplified by two Realistic 50 watt speakers (Model 14–1024). Emile subsequently reported that he was able to imagine Scenes 1, 2, and 3 with greater clarity. Within this basic framework each student's SUDS level was monitored at 2-min intervals and every complete series of probe assessments was followed by 5 min of relaxation exercises.

In vitro flooding. Each session was initiated by approximately 10 min of therapist-directed relaxation exercises. Sixty min of *in vitro* flooding were subsequently presented, wherein the students were asked to imagine the exact context of the designated scenes through the use of stimulus and response imagery cues. The 60-min flooding sessions were followed by 10 min of relaxation exercises. Each of the remaining scenes was subsequently presented for 6 min and probe assessments were conducted at 2-min intervals. Five min of relaxation exercises separated each series of 6-min probe assessments. Examined sequentially, Layla and Sawsan received two *in vitro* flooding sessions a week (i.e., two for each traumatic scene), and this schedule was maintained for 4 weeks. Emile's treatment followed a similar regimen with respect to Scenes 1, 2, and 3. Emile's fourth scene was more resistant to treatment and a third session was required.

Posttreatment and follow-up probes. Posttreatment and follow-up SUDS assessments were presented 3 days after the last treatment session and 6 months later according to the same regimen that was used during the baseline phase.

RESULTS

Affective Measures

The baseline SUDS probes reflected a high level of arousal with limited variation and a static trend. *In vitro* flooding was associated with a rapid and generally consistent reduction in the level of SUDS. Layla and Emile's posttreatment probes reflected virtually no discomfort. In a similar vein, Sawsan reported zero SUDS ratings for Scenes 1 and 2. Sawsan could not, however, totally overcome her discomfort with respect to Scenes 3 and 4, as indicated by posttreatment SUDS averages of 1 and 2 for the respective scenes.

Six-month follow-up SUDS averages revealed a notable decrease in the levels of arousal as compared to the baseline levels. Layla's 6-month follow-up SUDS averages for Scenes 1 through 4 were: 0.6, 1.0, 1.6, and 2.3. Sawsan's 6-month follow-up ratings consistently reflected zero SUDS ratings for Scenes 3 and 4. Her SUDS averages for Scenes 3 and 4 were 3 and 4. Emile's 6-month follow-up SUDS averages for Scenes 1 through 4 were 1.3, 0.6, 2.3, and 1.3. Figures 1, 2, and 3 provide a schematic summary of the SUDS data.

Layla's scores on the RCMAS (11-year-old female norm: M = 18.79, SD = 5.10, Saigh, 1985b) were 30 before the baseline probes, 20 after the last treatment session, and 18, 6 months later. Sawsan's scores were 27, 20, and 21. Emile's RCMAS scores (12-year-old male norm: M = 16.14, SD = 5.45, Saigh, 1985b) were 25, 18, and 16. Layla's CDI scores (11-year-old female norm: M = 15.91, SD = 6.22, Saigh, 1985b) were 23, 20, and 19. In a similar vein, Sawsan's scores were 25, 20, and 21. Emile's CDI scores (12-year-old male norm: M = 12.40, SD = 5.16) were 16, 13, and 12.

Figure 4 reflects the daily self-monitored frequencies of traumatic thoughts. It is evident that the prebaseline as well as the baseline ratings were fairly consistent for each student. *In vitro* flooding may have initially induced a moderate exacerbation of trauma-related recollections. Ongoing assessments did, however, reveal a reversal of the initial trend to the extent that trauma-related thoughts were virtually eliminated when the treatments were concluded. Six-month follow-up assessments were in keeping with the posttreatment levels and trends.

Behavioral Measures

The BATs were concurrently rated by each student's mother and by an MA level school psychologist. The level of agreement between the rates was determined by dividing the number of agreements by the number of disagreements and multiplying by 100. Rates of agreement for Layla's scores were 80%, 90%, and 100%. Sawsan's estimates were 80%, 90%, and 90%, and Emile's estimates were 80%, 90%, and 90%. Layla accomplished 35%, 95%, and 95% of her criteria during the respective assessments. Sawsan demonstrated 35%, 95%, and 95% of her criteria, and Emile completed 45%, 95%, and 95% of his criteria.

Layla's CTRS ratings (11-year-old female norm: M = 63.64, SD = 10.99, Saigh, 1985b) were 88, 83, and 70. Sawsan's CTRS scores were 90, 81, and 78. Emile's CTRS ratings (12-year-old male norm: M = 64.62, SD = 13.30, Saigh, 1985b) were 78, 68, 65.

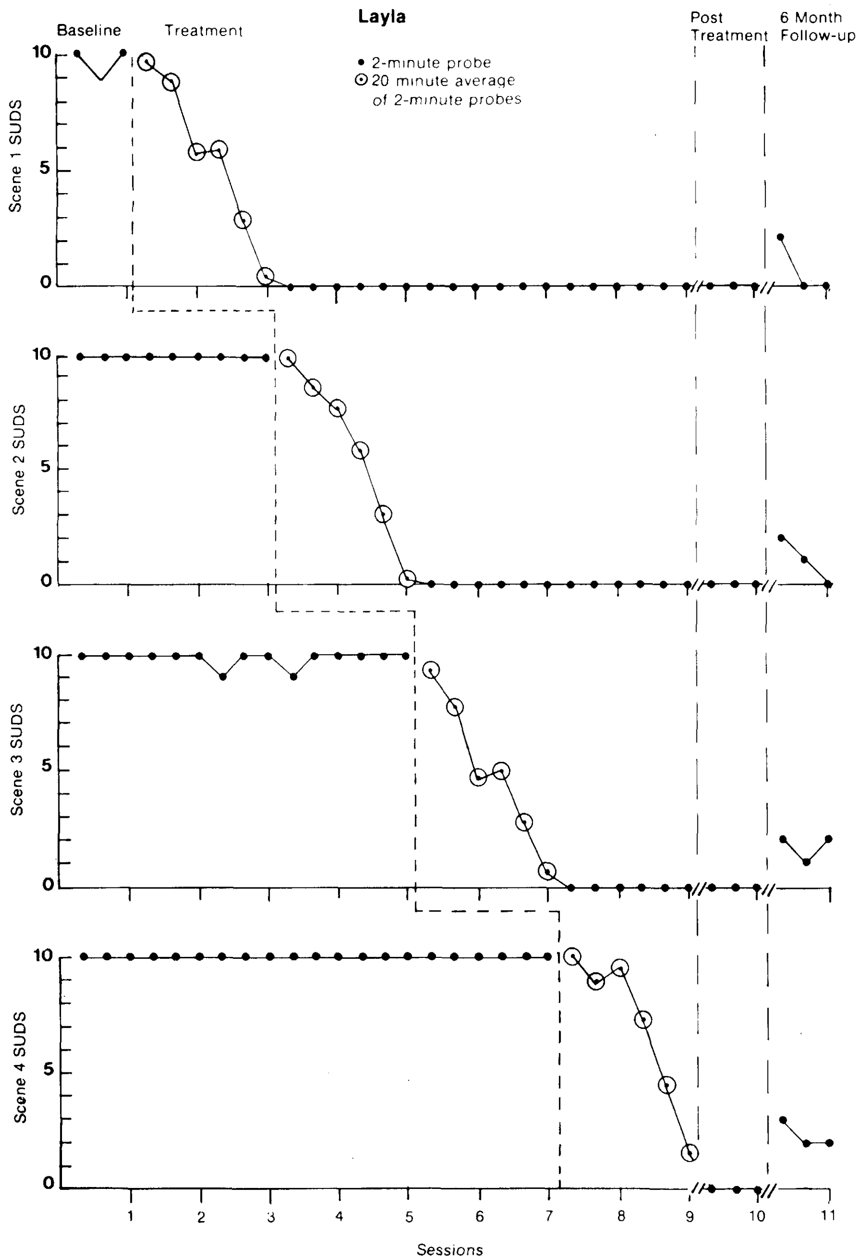

FIG. 1 Layla's SUDS level across sessions.

Cognitive Measures

Layla's Digit Span scores were 8, 10, and 11 and her Coding scores were 10, 12, and 12. Sawsan's Digit Span scores were 7, 10, and 10 and her Coding scores were 8, 9, and 9. Emile's Digit Span scores were 8, 10, and 11 and his Coding scores were 10, 10, and 11.

Layla's pretreatment GPA was 55%. Her GPA for the trimester that ended 10 weeks after the treatment was concluded was 65% and her GPA for the trimester that ended 6

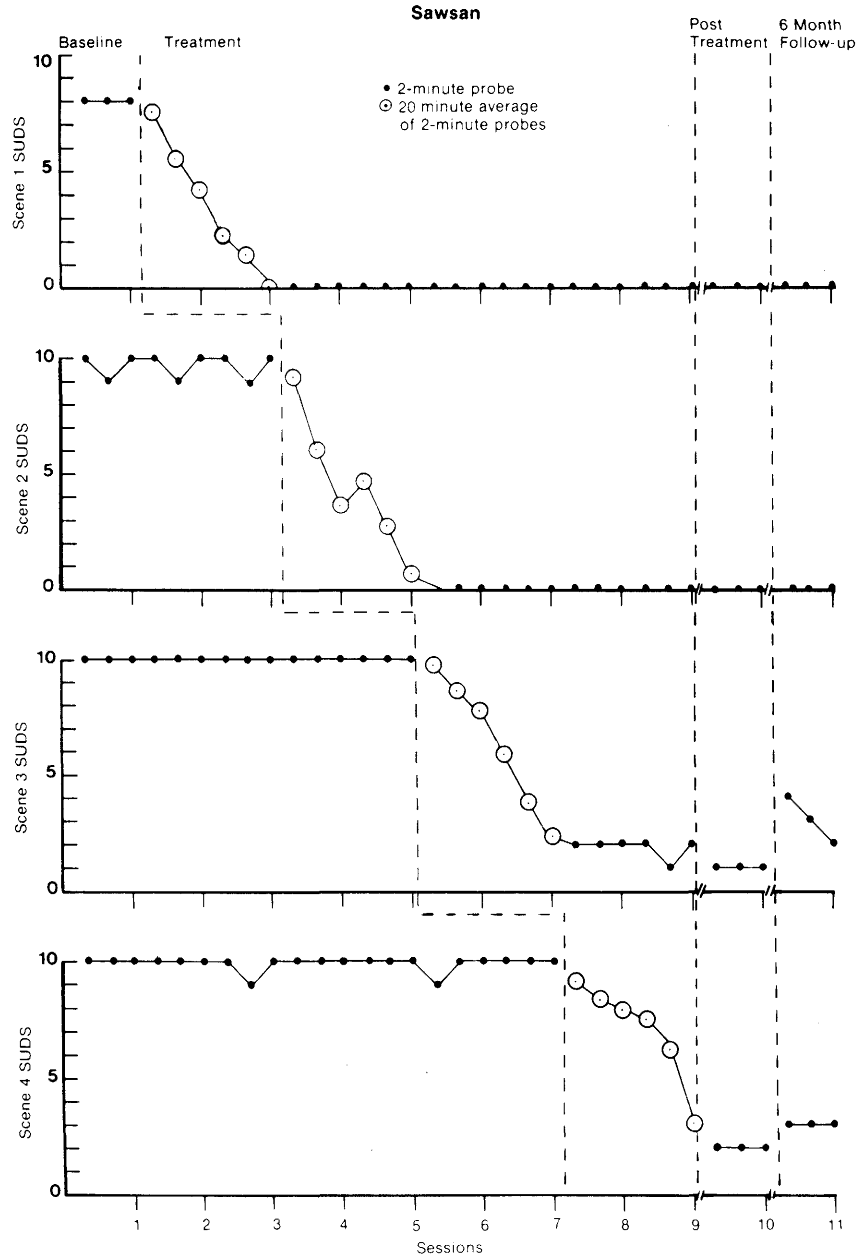

FIG. 2 Sawsan's SUDS level across sessions.

months after the last treatment session was 70%. Sawson's GPAs were 60%, 70%, and 70%, and Emile's were 50%, 60%, and 65%.

DISCUSSION

The results clearly suggest that the *in vitro* flooding package had a palliative influence on the students' affective, behavioral, and cognitive parameters. This observation assumes added significance when it is recalled that the students' symptoms had not

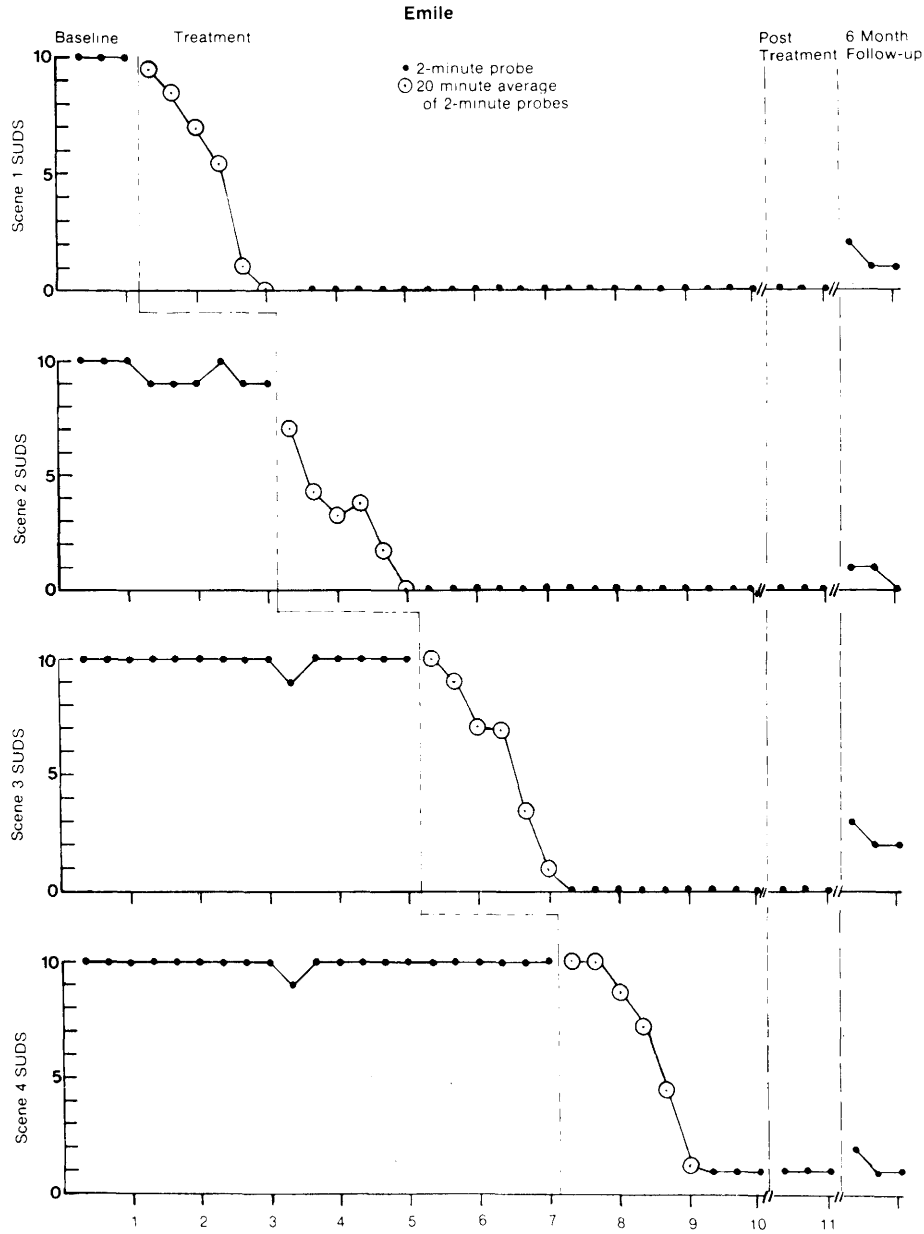

FIG. 3 Emile's SUDS level across sessions.

remitted for more than 2 years despite the fact that approximately two thirds of all neurotic disorders spontaneously remit within 2 years (Rachman & Wilson, 1980), and that marked improvements occurred after eight to nine treatment sessions.

The more prominent SUDS, traumatic thought, and BAT improvements that were registered at posttreatment and follow-up may be attributed to the primary focus of the treatment. More specifically, as the students' primary complaints involved trauma-related anxiety and as the *in vitro* flooding process directly focused on the reduction of traumatic anxiety, it follows that the treatment effects should have been more notably evinced through the measures that were directly associated with traumatic anxiety. It is also argued that the attenuated RCMAS and Digit Span scores are indicative of the fact

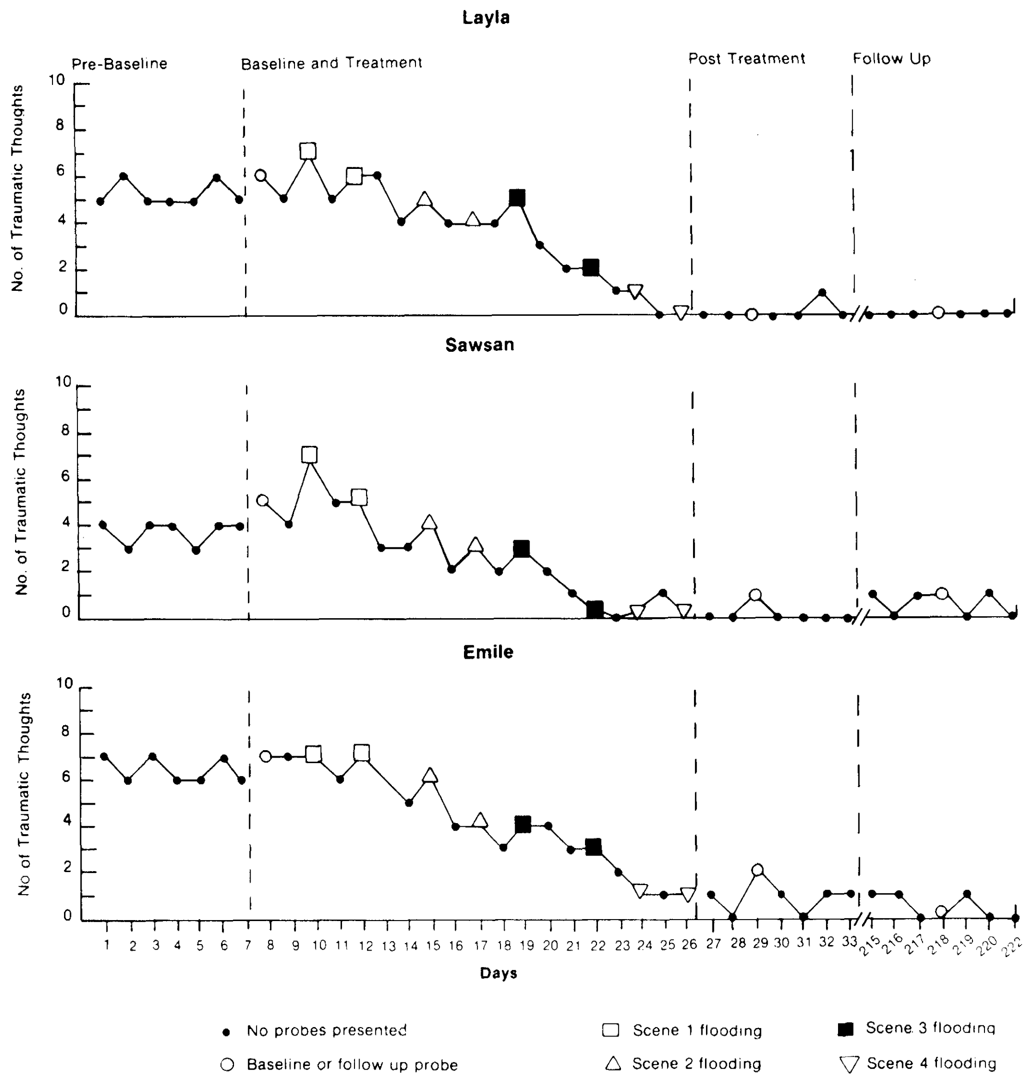

FIG. 4 Layla, Sawsan, and Emile's self-monitored frequencies of traumatic thoughts.

that these measures reflect traumatic anxiety to a lesser extent than the SUDS, traumatic thought, and BAT measures. It is also proposed that although the *in vitro* flooding package did not directly address the depressive, acting out, concentration impairment, or academic difficulties of the students (i.e., associated features of PTSD) (APA, 1980; Pynoos & Eth, 1985), the marked reduction in traumatic anxiety may have partially mitigated these problems as evinced by the modest but meaningful CDI, CTRS, Coding, and GPA improvements.

Although the overall results are promising, investigators are cautioned against formulating conclusions pertaining to the etiology of the disorder (e.g., PTSD is due to classical conditioning) because "one cannot argue that because a particular treatment is effective for a particular condition it necessarily throws light on the aetiology of that condition" (Marks, 1981, p. 8). Moreover, the results should be tempered with the realization that they were derived through the application of a multiple-component treatment package (i.e., relaxation, self-monitoring, and *in vitro* flooding) and that it is not possible to attribute the results to a single component of the overall package at this time.

The general efficacy of this procedure is concordant with the results that were reported in the United States when older PTSD cases received similar *in vitro* flooding packages (Fairbanks et al., 1983; Fairbank & Keane, 1982; Keane & Kaloupek, 1982). Perhaps more significantly, the observed results are in synchrony with the results that were observed when similar *in vitro* flooding packages were presented to traumatized youths in Lebanon (Saigh, 1985a, in press). It is interesting to note in this context that Saigh's earlier reports also evinced greater improvements on the outcome measures that were closely associated with traumatic anxiety. As such, behavioral investigators and practitioners may wish to add extra dimensions to the extent package such as self-instructional training (Meichenbaum, 1978) or daily report cards (O'Leary & Wilson, 1987) inasmuch as these techniques are more attuned to the treatment of the associated features of PTSD.

Examined from a different perspective, a number of considerations should be kept in mind. Initially, it should be understood that the students and their parents were fully appraised about the aversive nature of the flooding modality and that they selected *in vitro* flooding in lieu of systematic desensitization (i.e., a less aversive treatment). Moreover, the treatments as well as the posttreatment and follow-up assessments were presented during a protracted interval of relaxation in the level of hostilities in Lebanon. As such, Layla and Sawsan were not in danger when they carried out their behavioral walks. Analogously, although Emile experienced a considerable amount of discomfort by returning to, and remaining in, the room that was shelled, these actions were not dangerous activities when the study was conducted. Along these lines it is important to note that the students expressed considerable satisfaction with the treatment and that they indicated that the outcome was worth the momentary discomfort that they experienced during the flooding sessions. Finally, it should be realized that the treatment was directed toward reducing trauma-related anxiety and not situationally specific fears that may be logically warranted given the unrest in Lebanon.

Although this report involved the treatment of war-related PTSD cases, information pertaining to the assessment and treatment of PTSD is relevant to the practice of school psychology in America for a number of reasons. Recalling that PTSD may be induced by a "significant" stressor that is beyond the scope of normal human experience (APA, 1980) and that war stressors reflect a single category of "significant" stress, it follows that violent acts, natural disasters, and accidents may induce PTSD. It should be noted in this regard that 1,237,980 violent crimes (i.e., murder, aggravated assault, rape, and arson) were reported in the United States during 1984, and that approximately 29% of the victims of these crimes were below the age of 19 (U.S. Department of Justice, 1985). Although information pertaining to the incidence of PTSD following exposure to puissant stressors varies (Saigh, 1984, 1985c), PTSD cases are evident in American schools (Pynoos & Eth, 1985). In view of this and the modicum of information pertaining to the assessment and treatment of childhood PTSD, it is felt that the call to pursue the search for a validated treatment of choice is in keeping with the interests of the profession.

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